SCHOOL OF MATH

CLASS 2

Multiplication Facts

×3, ×4, ×6

Patterns • Strategies • Real-World Problems



×3:
Double, then add once more.

×4:
Double, then double again.

×6:
Multiply by 3, then double.

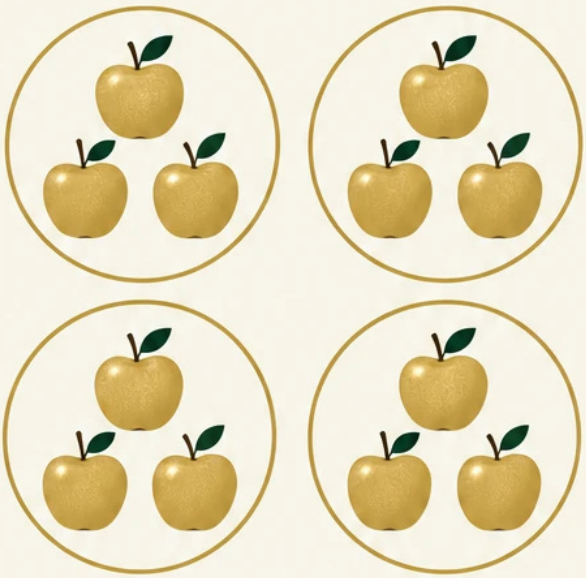
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CLASS 2

Multiplication Facts ×3, ×4, ×6

1. Equal Groups and Arrays

A. Equal groups



4 groups of 3

3 + 3 + 3 + 3 = 12

B. An array



4 rows × 3 columns

4 × 3 = 12

Equal groups have the same number in each group.

Image 1 practice: Equal groups: name the factors

1.1. There are 5 bowls with 3 grapes in each. Write a multiplication equation and a repeated-addition equation.

1.2. Four boxes each hold 5 blocks. Name both factors and the product.

1.3. Rewrite 6 + 6 + 6 as a multiplication equation using number of groups first.

1.4. Three bags contain 2, 3, and 4 stones. Can you use 3 × 3 to describe three equal bags? Explain and find the total.

1.5. A tray holds 4 cookies. How many cookies are on 5 full trays?

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2. Skip Counting by 3, 4, and 6

Add the same number each time.

Count by 3

3	6	9	12	15
18	21	24	27	30

Each step adds 3.

4 steps:
$4 \times 3 = 12$

Count by 4

4	8	12	16	20
24	28	32	36	40

Each step adds 4.

6 steps:
$6 \times 4 = 24$

Count by 6

6	12	18	24	30
36	42	48	54	60

Each step adds 6.

7 steps:
$7 \times 6 = 42$

Your turn

3, 6, 9, ___, ___

6, 12, 18, ___, ___

Image 2 practice: Skip counting by 3, 4, and 6

2.1. Continue the pattern: 15, 18, 21, ___, ___. State the size of each step.

2.2. Fill the gaps while counting by 4: 8, 12, ___, 20, ___.

2.3. Start at 0 and make 6 jumps of 6. List the landing numbers and write a multiplication equation.

2.4. Ben lists 0, 4, 8, 12 and says, 'Four jumps of 4 land on 12.' Explain his mistake and find the landing number after four jumps.

2.5. A game token starts on 6 and moves forward 3 spaces on each turn. Where will it land after 5 turns? Show how you know.

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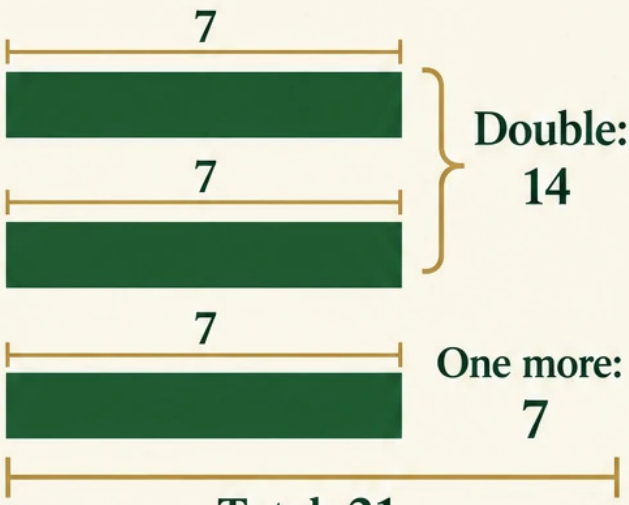
CLASS 2

Multiplication Facts ×3, ×4, ×6

3. Multiplying by 3

Double the number, then add it once more.

$$3 \times 7 = 21$$



1. Start with 7.
2. Double: $7 + 7 = 14$
3. Add one more 7: $14 + 7 = 21$

Try another fact.

$$3 \times 9 = (2 \times 9) + 9$$
$$= 18 + 9 = 27$$

×3 = ×2 + one more group

Image 3 practice: Multiply by 3: double and add

3.1. Use double-and-add to find 3 × 2. Show the double and the extra group.

3.2. Use double-and-add to find 3 × 11.

3.3. Fill both blanks: 3 × 5 = (2 × 5) + ___ = ___.

3.4. Mia says 3 × 6 = 12 + 1 = 13. What should she add instead, and why?

3.5. You know 3 × 10 = 30. Find 3 × 12 by adding two more groups of 3.

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CLASS 2 • Multiplication Facts ×3, ×4, ×6

3. Multiplying by 3

Double the number, then add it once more.

1
Start with 8.
8

2
Double it:
8 + 8 = 16

3
Add one more 8:
16 + 8 = 24



3 × 8 = 24

Another example:

3 × 6 = (2 × 6) + 6 = 12 + 6 = 18

Three groups = two groups + one more group.

Image 4 practice: Connect ×3 to ×4: add a group or double twice

4.1. Find 4 × 3 by doubling twice.

4.2. Find 4 × 9 by doubling twice.

4.3. You know 3 × 7 = 21. Find 4 × 7 without starting over.

4.4. Leo doubles 6 to get 12, then adds 2 to get 14 for 4 × 6. Correct his second step.

4.5. Find the missing factor: 4 × ___ = 44. Explain with doubles.

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5. Multiplying by 6

Multiply by 3, then double.

1

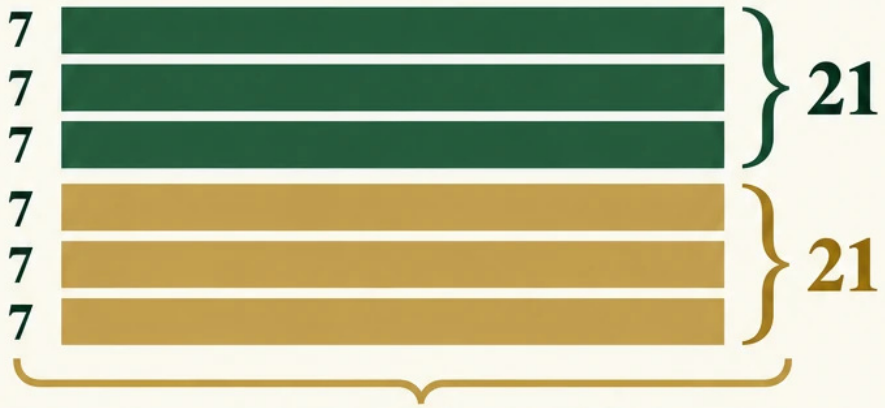
Step 1: Triple

$3 \times 7 = 21$

2

Step 2: Double

$2 \times 21 = 42$



$6 \times 7 = 42$

Another example:

$$6 \times 8 = 2 \times (3 \times 8) = 2 \times 24 = 48$$

Six groups = double three groups.

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Image 5 practice: Multiply by 6: double three groups

5.1. Find 6 × 3 by first finding 3 × 3.

5.2. Find 6 × 5 using triple-then-double.

5.3. If 3 × 11 = 33, what is 6 × 11?

5.4. Nora says 6 × 4 is 3 × 4 plus 3. Find and fix her mistake.

5.5. Find the missing factor: 6 × ___ = 60. Check with a ×3 fact.

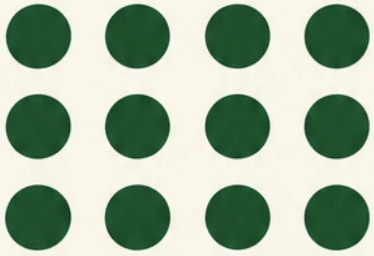
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6. Switching Factors

Changing the order keeps the same product.

3×4



3 rows of 4

$3 \times 4 = 12$

4×3



4 rows of 3

$4 \times 3 = 12$

$$3 \times 4 = 4 \times 3 = 12$$

The same 12 dots, arranged differently.

The commutative property

$$a \times b = b \times a$$

$$6 \times 4 = 4 \times 6 = 24$$

Know one fact? You also know its reverse!

Image 6 practice: Switching factors: the same product

6.1. Use 4 × 8 = 32 to find 8 × 4. Name the property.

6.2. Complete the switched pair: 6 × 9 = ___ × 6 = ___.

6.3. True or false: 3 × 5 = 5 × 3 means that both arrays have the same number of rows.

6.4. Which equation shows switching factors? A: 4 × 5 = 5 × 4. B: 4 × 5 = 2 × 10. Explain.

6.5. Do 8 − 3 and 3 − 8 give the same answer? What does this tell you about switching?

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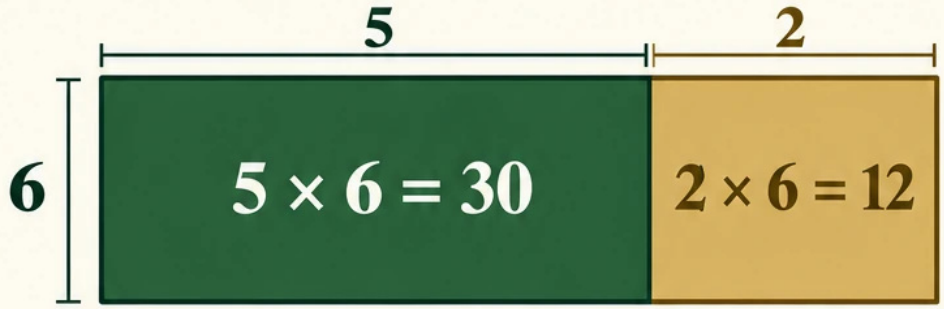
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7. Breaking Apart Tricky Facts

Use facts you know to find a new product.

7×6

Break 7 into 5 + 2.



$$7 \times 6 = (5 \times 6) + (2 \times 6)$$

$$30 + 12 = 42$$

$$7 \times 6 = 42$$

$$9 \times 4 = (5 \times 4) + (4 \times 4)$$
$$= 20 + 16 = 36$$

The distributive property

$$a \times (b + c) = a \times b + a \times c$$

Break apart. Multiply each part. Add the products.

Image 7 practice: Break apart a fact you do not know yet

7.1. Split 8 into 5 + 3 to find 8 × 3.

7.2. Split 11 into 10 + 1 to find 11 × 4.

7.3. Find 8 × 6 by splitting 8 into 4 + 4.

7.4. A student writes 6 × 12 = (6 × 10) + 2. Correct the equation.

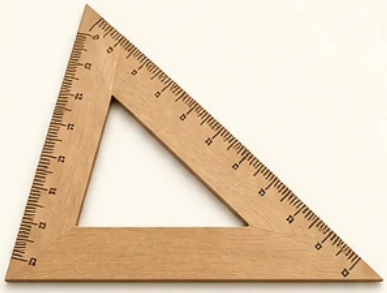
7.5. Find 9 × 3 using 10 × 3 minus one group of 3.

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8. Real-World Word Problems

Find equal groups. Multiply to find the total.



Triangles

A student draws 8 separate triangles. Each has 3 sides. How many sides in total?

8 × 3 = 24 sides



Pencil packs

Each pack holds 4 pencils. How many pencils are in 7 packs?

7 × 4 = 28 pencils



Egg cartons

Each carton holds 6 eggs. How many eggs are in 9 cartons?

9 × 6 = 54 eggs

Number of groups × Number in each group = Total

Your turn

A museum has 6 rows of seats with 4 seats in each row. How many seats are there?

6 × 4 = _____ seats

Image 8 practice: Use multiplication in a story

8.1. Four shelves each hold 9 books. How many books are there? Show a switched equation too.

8.2. Five tables each have 6 seats. How many seats are available?

8.3. A baker fills 3 trays with 11 rolls each, then sells 5 rolls. How many remain?

8.4. One display has 4 rows of 8 badges. Another has 8 rows of 4 badges. Which has more, or are they equal?

8.5. A class needs 25 paintbrushes. Each pack holds 6. Are 4 packs enough? If not, how many brushes are missing?

Homework review: R1–R5

R1. Review Class 1: Find 2 × 8 using doubles.

R2. Review Class 1: Find 5 × 7 by counting in fives.

R3. Review Class 1: A box holds 10 cards. How many cards are in 12 boxes?

R4. Review Class 1: If 8 × 5 = 40, find 8 × 10.

R5. Draw or describe two arrays showing 3 × 5 = 5 × 3.

Homework review: R6–R8

R6. Find 4 × 12 by doubling twice.

R7. Use 3 × 9 = 27 to find 6 × 9.

R8. Explain why 4 × 6 = 6 × 4 but 4 × 6 is not 4 + 6.

Homework word problems: W1–W5

W1. Three baskets each hold 5 pears. How many pears are there? Write the reversed fact too.

W2. Four children each make 6 paper stars. How many stars do they make altogether?

W3. A tray has 6 rows with 5 seed pots in each row. How many pots are there?

W4. A shop packs 3 erasers in each bag. How many erasers are in 11 bags?

W5. A display holds 4 rows of 10 postcards. It is rearranged into 10 equal rows using every postcard. How many postcards are in each new row?

Homework word problems: W6–W10

W6. Six teams each receive 4 bibs. Two spare bibs are added. How many bibs are there in total?

W7. Four boxes each contain 11 beads. A student uses 9 beads. How many remain?

W8. One club sets out 3 tables with 8 places each. Another sets out 4 tables with 6 places each. Do the clubs have the same number of places? Is this a switched pair?

W9. A library displays 8 rows of 3 books. Use 5 rows and 3 rows to find the total.

W10. A class needs 30 counters. Each packet contains 4 counters. Are 7 packets enough, and how many full packets are needed?

Homework word problems: W11–W12

W11. A florist makes 6 bunches with 9 flowers in each. She sells 2 whole bunches. How many flowers remain?

W12. A museum has 5 rows of 6 seats. A guide says rearranging them into 6 rows of 5 will add one seat. Explain the mistake and state the total.